Syllabus

Class Schedule: Lecture: T R: 2:00 PM – 3:20 PM (Section 01) – ISC 3342

Instructor:
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Office Hours: T W R 4:30 – 5:30 pm, and by appointment.

Course Description: This course introduces students to the principles of empirical inquiry, statistical modeling, and computational experimentation in data-rich and AI-supported environments. Leveraging modern tools such as Python and generative AI, students will explore how to design studies, quantify uncertainty, and perform model-driven inference using techniques foundational to machine learning, reinforcement learning, and causal reasoning. Topics include research design, simulation-based inference, likelihood-based estimation, confidence intervals, model comparison (Akike Information Criterion [AIC], Bayesian Information Criterion [BIC] and regularization techniques), Bayesian updating, and the use of upper confidence bounds for decision-making. While formal mathematical prerequisites are not required, students will be expected to engage in rigorous reasoning with the support of AI-assisted coding and explanation tools. The course emphasizes reproducible workflows and theory-grounded applications.

Course Objectives:

1. Provide students with a critical understanding of the essential concepts, theories, and techniques in statistics and the use of data.
2. Develop students' ability to apply effective methodologies and interpret the results.
3. Expose students to real-world problems that are being engaged with by contemporary problem-solvers and decision-makers.

Evaluation:

Class Participation	10%
Homework/Quizzes	50%
Midterm	20%
Final	20%

Letter Grades:

93 – 100 %	A	73 – 76.99%	C
90 – 92.99%	A-	70 – 72.99%	C-
87 – 89.99%	B+	67 – 69.99%	D+
83 – 86.99%	B	63 – 66.99%	D
80 – 82.99%	B-	60 – 62.99%	D-
77 – 79.99%	C+	0.0 – 59.99%	F

Resources: For Python programming, you need access to reliable cloud computing, such as Google COLAB by using the education link. **Please sign up for Colab ASAP,** in preparation for Python coding. We use Piazza for discussions and class participation. I aim to deliver the study material in a self-contained and efficient manner, so no expensive textbook is necessary. In this course, you will engage with fundamental, conceptual ideas and applications.

Class Participation: You will receive full credit for participation ONLY if you actively engage in class discussions, contribute to coding sessions, and provide feedback to your peers, including Piazza posts. This is not a guaranteed 100%; in Spring 2025, the average was ~70%. An average of 20 minutes per class will be used for group discussions and active coding. You can work in small groups to understand and debate the class materials, exercises, and the data and code applications; the instructor mediates all the class discussions. Drafts of the code for solving the problems are shared and discussed in class in a round table format; the instructor and all participants in the debate provide feedback.

Homework: The homework assignments will be posted on Blackboard at least five days before the due date. The expectation is that you will work on the assignment as the material is covered, so do not be surprised if the due dates are relatively close to when we finish covering the material. If you have any questions, please actively engage in Q/As on Piazza. The grading of homework questions also involves partial credit. The lowest homework or quiz score will be dropped to accommodate an unexpected personal problem, short-term illness, or class absences. **Late submissions** are possible only with a **penalty** adjustment communicated by the **assignment's instructions**.

Quizzes: Quizzes will be on paper in-class, and involve short-answer, multiple-choice, and/or multiple-answer questions, as well as explaining concepts or the solving of problems provided to you (optionally using logical diagrams) to determine the answer. You will be allowed a four-operations calculator when needed. You will be allowed to bring both sides of one **handwritten** 4"x6" notecard for each quiz with any notes that you want. You may **not** print the content of your notecard out for any quiz. The lowest quiz or homework score will be dropped to accommodate an unexpected personal problem, short-term illness, or class absences. Quizzes and homework will be weighted equally.

Midterm and Final: A midterm (test: October 6th, project: October 15th) and a final (December 10th) will have both in-class closed-book conceptual questions and take-home project parts. They assess your knowledge of the foundational concepts of research design, probability and statistics presented during the course.

Important Dates: Please read the important dates and plan accordingly.

- The add-and-drop deadline is September 4th.
- Fall Break (no classes) October 8th – 11th.
- Midterm Test - October 6th.
- Midterm Project – October 15th.
- The withdrawal deadline is October 26th.
- Thanksgiving Break (no classes) November 25th - 29th.
- The last day of classes is December 4th.
- Final Exam - December 10th between 2 pm – 5 pm.

Textbooks(optional):

1. ***The Effect: An Introduction to Research Design and Causality*** by Nick Huntington-Klein
2. ***An Introduction to Statistical Learning with Applications in Python*** by G. James, D. Witten, T. Hastie, R. Tibshirani, J. Taylor
3. ***Probability & Statistics for Engineers & Scientists*** by R.E. Walpole, R.H. Myers, Keyin Ye
4. ***Practical Statistics for Data Scientists*** by P. Bruce, A. Bruce, P. Gedeck

Learning Modules:

Module 1: Foundations of Research, Causal Reasoning, and Workflows

Goal Introduce students to the structure of empirical inquiry, emphasizing reproducible workflows, causal reasoning, and the distinction between experimental and observational data.

Topics

- **Structure of Inquiry:** What is research design?
- **Causal Reasoning:** Correlation vs. Causation, identifying confounders, and the basics of causal inference.
- **Data Types:** Continuous (interval or ratio level of measurement), ordinal, nominal.
- **Variables:** Independent, dependent, controls, and confounders.
- **Study Types:** Observational vs. experimental studies.
- **Reproducibility:** Setting up reproducible workflows (e.g., version control, organized notebooks) for collaborative science.
- **Sampling:** Sampling bias and error.
- **AI Integration:** Using AI to refine research questions, summarize literature, and critique experimental designs.

Applications

- Critique a flawed study using an AI assistant, specifically looking for causal fallacies.
- Design a small observational study and use AI to assist with refinement and variable selection.

Module 2: Descriptive Statistics, Visualization, and Exploration

Goal Help students understand how to summarize and visualize data meaningfully, identifying patterns while avoiding common pitfalls in interpretation.

Topics

- **Numerical Summaries:** Mean, median, variance, percentiles.
- **Visual Summaries:** Histograms, boxplots, bar charts, scatterplots.
- **Relationships:** Correlation and simple linear regression.
- **Errors in Logic:** Misuse of summaries (e.g., calculating means of nominal variables) and overfitting concepts.
- **Computational Tools:** Using Python and AI to explore and visualize datasets.

Applications

- Use Python to analyze a small dataset (e.g., survey results) and generate reproducible plots.
- Use AI to critique poor visualizations or improve them for clarity.

Module 3: Probability, Distributions, and Bayesian Thinking

Goal Build foundational understanding of uncertainty, probability, and Bayesian updating, connecting theoretical distributions to real-world data.

Topics

- **Foundations:** Sample spaces and definition of probability.
- **Counting:** Permutations and combinations.
- **Distributions:** Discrete (binomial, Poisson) and Continuous (introduction to normal distribution and CLT).
- **Expectation:** Expected value and variance (conceptually).

- **Bayesian Reasoning:** Conditional probability and Bayesian updating (updating beliefs based on new evidence).
- **Simulations:** Show normality emerging from binomial via simulations; reinforce with non-normal distributions (e.g., beta).
- **AI Support:** Using AI to identify appropriate distributions for specific data types.

Applications

- Simulate a binomial process using Python and visualize the results.
- Perform a simple Bayesian update calculation using Python to adjust probabilities after an observed event.

Module 4: Inference, Model Comparison, and Estimation

Goal Teach students to reason from samples to populations using likelihood-based estimation, simulation, and rigorous model comparison techniques.

Topics

- **Inference Basics:** Populations vs. samples; empirical distributions and confidence intervals.
- **Hypothesis Testing:** Null vs. alternative hypotheses; p-values; Type I and Type II errors.
- **Estimation:** Likelihood-based estimation and Maximum Likelihood Estimation (MLE) concepts.
- **Model Comparison:** Comparing models using Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC).
- **Complexity Control:** Introduction to regularization techniques (Lasso/Ridge concepts) to prevent overfitting.
- **Simulation-Based Inference:** Permutation tests and bootstrapping.
- **Classic Tests:** T-test, chi-squared, ANOVA.

Applications

- Compare two sample groups using both a permutation test and a standard t-test.
- Fit two competing models to a dataset in Python and use AIC/BIC to select the best one.

Module 5: Decision Making and Integrated Analysis

Goal Integrate prior knowledge by designing a study that utilizes decision-making frameworks (like UCB) and ethical considerations, leveraging AI and Python.

Topics

- **Decision Making:** Upper Confidence Bounds (UCB) for decision-making and exploration vs. exploitation (Reinforcement Learning foundations).
- **Project Planning:** End-to-end planning from research question to conclusion.
- **AI Assistance:** Using AI for study design, exploratory analysis, and model checking.
- **Ethics:** Ethical considerations, transparency, bias, and limitations in AI-supported analysis.
- **Interpretation:** Interpreting results in context.

Applications

- **Project:** Students develop a mini research project (individually or in groups) that includes data analysis, model selection (or UCB implementation), and results presentation.
- Use AI to generate critiques and suggest improvements to peer projects.

Weekly Schedule:

Week	What we will discuss/do in class	What to read before class
Week 1	Foundations of research and causal reasoning	From https://theeffectbook.net/ Chapters 1 – 4
Week 2	Causal workflows Descriptive Statistics	From https://theeffectbook.net/ Chapters 5 – 8
Week 3	Descriptive Statistics, Visualization and Exploration	Iterators, operations with NumPy arrays, and Matplotlib plots.
Week 4	Probability and Bayesian Reasoning	Law of Large Numbers, Conditional Event and Monte Carlo simulations
Week 5	Random Variables and Distributions	The Law of Probable Errors and the Central Limit Theorem
Week 6	Midterm Test (October 6th)	Review the first three modules.
	Fall Break (October 8 – 11)	
Week 7	Midterm Project (October 15th)	
Week 8	Inference and hypothesis testing	Hypothesis testing, normality tests
Week 9	Maximum Likelihood and Estimation	Confidence intervals, Chernoff Bounds, BLUE estimators
Week 10	Model Comparison and Selection	Spurious correlations, The rank of a matrix, under-determined systems
Week 11	Decision Making and Integrated Analysis	
Week 12	Decision Making and Integrated Analysis Applications	

Due Dates:

Quiz 1	September 3rd
Quiz 2	September 10th
HW 1	September 17th
Quiz 3	September 24th
Midterm Test	October 6th
Midterm Project	October 15th
Quiz 4	October 22nd
HW 2	October 27th
Quiz 5	November 5th
Quiz 6	November 12th
HW 3	November 19th
Final	December 10th

Collaboration: Collaboration is both allowed and encouraged. Most of the time, we do not work in isolation, and using the resources available is both wise and efficient. You are welcome to work collaboratively with classmates – that means you can work together to solve problems together…not that one person does the work and others copy it. You are also welcome to use the internet as a resource to refresh your memory, clarify concepts, and help with short code snippets. You must cite your sources, and you should not be copying anything wholesale from any source: internet, human, artificial intelligence, or otherwise.

Things that are off limits:

- Soliciting (hey, what did you get for question X? Can you send me your code?) and copying code and answers from another person
- Copying code/answers directly from internet sources
- Using sources appropriately but failing to cite them
- Getting generative AI to do your work for you

Think about it: do you want to do all of the work and then give the answers to someone? Is that fair to you? Do you wholeheartedly trust that the answers you get from sources (ahem, ChatGPT) will be 100% correct? Do you want to attend a job interview and explain that you don't understand the material? Do you know the answers you are deriving from external sources well enough that you could explain them when asked?
You should always be ready and able to defend every answer you submit confidently. This means that if I ask you to meet with me to explain any answer you submit, you should be able to do so in detail yourself.

Possible Changes: Any changes to the following course schedule or due dates will be announced in class and on Blackboard ahead of time.

Guidance: Students performing at a C level or below must schedule a meeting with the instructor to discuss class performance.

Class Absences: If you miss a class or two, check the material covered in class posted on Blackboard. For additional help, you can make an appointment with the TA or the instructor. Long-term absences need to be documented via the Dean of Students Office.

Honor Code: Academic integrity is at the heart of the university, and we are all responsible for upholding the ideals of honor and integrity. The student-led honor system is responsible for resolving any suspected violations of the Honor Code, and I will report all suspected instances of academic dishonesty to the honor system. The Student Handbook (www.wm.edu/studenthandbook) includes your responsibilities as a student. Your full participation and observance of the Honor Code are expected. To read the Honor Code, see www.wm.edu/honor.

Artificial Intelligence: The use or incorporation of AI-generated content (from ChatGPT, Gemini, Co-Pilot, etc.) is permitted only in certain designated assignments and setups. For in-class quizzes or tests it is strictly prohibited.

Student Accessibility: William & Mary accommodates students with disabilities in accordance with federal laws and university policy. Any student who feels they may need an accommodation based on the impact of a learning, psychiatric, physical, or chronic health diagnosis should contact Student Accessibility Services staff at 757-221-2512 or sas@wm.edu to determine if accommodations are warranted and to obtain an official letter of accommodation. For more information, please see www.wm.edu/sas.

Tech Support Center: Email: Support@wm.edu
Phone: 757-221-HELP
Walk in Support: Jones Hall, Room 201, 200 Ukrop Way
Hours: Monday- Friday 8am-5pm

Student Success:

Student Success supports students' personal growth, development, engagement, and belonging by providing holistic guidance as they navigate their W&M journey. It encompasses the offices of Academic Wellbeing, Care Support Services, and Student Accessibility Services.

- For academic support such as tutoring, time management, study skills, and educational coaching, please contact Academic Wellbeing at wm.edu/academicwellbeing (academicwellbeing@wm.edu)
- For concerns about the well-being of a member of the William & Mary community or to seek assistance for interpersonal, academic, and wellness challenges, please contact Care Support Services at wm.edu/care (care@wm.edu)
- For accommodation needs or questions, please contact Student Accessibility Services at wm.edu/sas (sas@wm.edu)

Student Health:

William & Mary recognizes that students juggle different responsibilities and can face challenges that make learning difficult. There are many resources available at W&M to help students navigate emotional/psychological, physical/medical, material/accessibility concerns, including:

- The W&M Counseling Center at (757) 221-3620. Services are free and confidential.
- The W&M Health Center at (757) 221-4386.
- To seek assistance for interpersonal, academic, and wellness challenges, please contact Care Support Services at wm.edu/care (care@wm.edu).
- For a list of other resources available to students, see here or:

For psychological/emotional stress, contact the Counseling Center by email or 757-221-3620.
For physical/medical concerns, contact the Health Center by email or at 757-221-4386.
Both are located at 240 Gooch Drive. Services are free and confidential.
If you or someone you know needs additional support or resources, please contact the Dean of Students by submitting a care report online, by phone at 757-221-2510, or by email.

Expectations:

All students are strongly encouraged to bring their laptops or notebook computers to class since we plan on doing a significant amount of active coding.
We will address instructor absence/illness by scheduling remote (online) classes for no more than a week. In extreme cases, a substitute instructor will be arranged.

IT Resources:

Information Technology for Students
Service Status-IT -This is the service status page for William & Mary. Members of the W&M community are invited to subscribe to receive updates for services they use regularly. Some services have established monthly maintenance windows where complete or intermittent service interruptions occur.
Get Software at W&M-The Information Technology Software site offers a collection of free and discounted software for the William & Mary community. Software such as Microsoft Office and Adobe products are available for purchase at a **discounted rate.**
Microsoft Office 365, Student Email- O365
Blackboard Help
Getting Started Playlist for Ultra Courses
Blackboard Ally- Alternative File Formats
Blackboard Mobile App